

ASFI AHAMED

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Profile

Computer Science and Engineering undergraduate with hands-on experience building and deploying software and machine-learning applications, with interests in software engineering, data science, and reliable real-world systems.

Education

B.Sc. Engineering (Hons) in Computer Science & Engineering, University of Moratuwa	2024 – Present
CGPA: 3.39/4.00 (through Semester 4)	Moratuwa, Sri Lanka
G.C.E. Advanced Level, KM/Al-Ashraq National School	2022
3A Z-Score: 2.3250 Island Rank: 424	Nintavur, Sri Lanka

Selected Projects

NovaHub – Real-Time Workspace Collaboration Platform [Live App](#) | [GitHub](#) **2026 – Present**

- Built and deployed a full-stack collaboration platform with JWT authentication, workspace authorization, persistent messaging, and authenticated Socket.IO rooms.
 - Implemented live member updates and automatic reconnect recovery with missed-message synchronization after temporary network interruptions.
- Tech: React, Node.js, Express, MongoDB, Socket.IO, Tailwind CSS, Vercel, Railway*

Personal Portfolio & Custom CMS [Website](#) | [Demo](#) | [GitHub](#) **2026**

- Built a protected custom CMS for managing portfolio projects, skills, education, experience, certificates, social links, branding, and other public-site content.
 - Implemented reusable CRUD workflows, Supabase authentication, PostgreSQL RLS, storage, configurable ordering, and appreciation analytics.
- Tech: React, TypeScript, Supabase, PostgreSQL, Tailwind CSS, Vite, Vercel*

ViewCastLK – YouTube View Count Forecasting *Team Project* **2026 – Present**

- Contributed to a leakage-aware multi-horizon forecasting system for predicting cumulative YouTube views at Day 7, 14, 21, and 30 using pre-publication metadata and historical channel context.
 - Prototyped quota-aware YouTube API collection, built the forecasting dashboard, and implemented automated backups for the Supabase/PostgreSQL data pipeline.
- Tech: Python, Pandas, YouTube Data API v3, PostgreSQL/Supabase, GitHub Actions, Next.js, React, TypeScript, Recharts*

Student Performance Predictor [Live App](#) | [GitHub](#) **2026**

- Built an end-to-end regression pipeline with numerical/categorical preprocessing and model comparison for student math-score prediction.
 - Applied GridSearchCV, persisted preprocessing/model artifacts, and deployed Flask-based inference on AWS Elastic Beanstalk.
- Tech: Python, Pandas, scikit-learn, XGBoost, CatBoost, Flask, AWS*

Book Recommender System [Live App](#) | [GitHub](#) **2026**

- Built and deployed an item-based collaborative-filtering recommender using explicit Book-Crossing ratings, sparse CSR features, and cosine nearest-neighbor retrieval.
 - Engineered reproducible ingestion/training, model-artifact validation, automated tests, and Docker-based inference packaging.
- Tech: Python, Pandas, NumPy, SciPy, scikit-learn, Streamlit, pytest, Docker*

NoteBot – PDF RAG Study Assistant [Live App](#) | [GitHub](#) **2026**

- Engineered a page-aware PDF RAG pipeline using embeddings and custom NumPy cosine-similarity retrieval.
 - Integrated Qwen, OpenAI, and Cloudflare Workers AI with provider controls, safe response handling, and 30 passing unit/contract tests.
- Tech: Python, Streamlit, NumPy, FastEmbed, Qwen, OpenAI API, Cloudflare Workers AI*

Research & Publication

Environmental Drivers of Respiratory Disease: A District Level Analysis [Paper](#) | [Dataset](#) | [GitHub](#) **2026** *Accepted at MERCon 2026*

- Contributed to a team environmental-health study covering all 25 districts of Sri Lanka from 2014–2024 using multi-source environmental and health data.
- Contributed to temporally validated XGBoost and SHAP analysis and the release of the accompanying reproducible public dataset.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C++, SQL

Software Development: React, Node.js, Express, Flask, Socket.IO, Tailwind CSS

Databases: PostgreSQL, MongoDB, MySQL, Supabase

Data Science & ML: Pandas, NumPy, SciPy, scikit-learn, XGBoost, SHAP

AI & Applications: RAG, Embeddings, Streamlit, OpenAI API

Tools & Cloud: Git, GitHub, Docker, AWS, Vercel, Railway, GitHub Actions

Certifications

AWS Academy: Cloud Foundations; Microservices and CI/CD Pipeline Builder (2026)

NVIDIA: Fundamentals of Accelerated Data Science (2025)

Kaggle Learn: Python, Pandas, Intro to Machine Learning, Data Cleaning, Feature Engineering (2026)

Generative AI: Generative AI for Beginners – Build AI Chatbot, Jennys Lectures (2026)

Data Analysis: Data Xplore 2.0, Statistics Society – University of Sri Jayewardenepura (2026)

Coding Competitions: Innovate with Ballerina – IEEE UoM & WSO2; MoraXtreme 10.0 – IEEE UoM (2025)

Activities

Company Coordinator – Career Fair Day, University of Moratuwa (2026) | **OC Logistics Member** – SLIoT | **Logistics Support** – Sakura Event